Stock Sentinel – Insight Report

INTRODUCTION:

This project analyzes stock market data using technical indicators and machine learning techniques. The objective is to identify trends, generate trading signals, and assist in decision-making. Indicators such as Moving Averages, RSI, Bollinger Bands, and volatility were used along with a Random Forest model for prediction.

INSIGHT:

Among the selected stocks, **HDFCBANK.NS** showed relatively stable movement and better momentum compared to others.

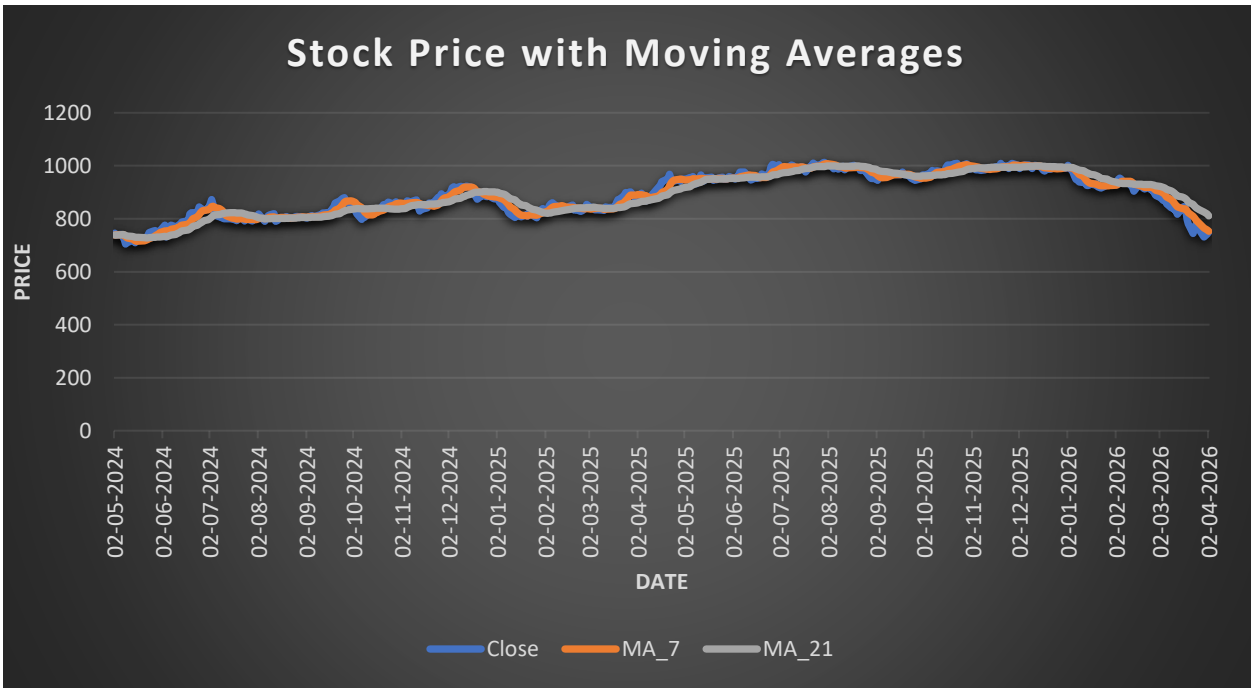
Among the analyzed stocks, **HDFCBANK.NS** showed the strongest momentum. This is evident from its consistent price trend, smoother moving averages, and relatively stable RSI levels. The stock maintained a balanced movement without extreme volatility, indicating a more reliable trend.

ML MODEL PERFORMANCE:

The Random Forest machine learning model was used to predict stock movement. The model achieved an accuracy of approximately **53%**, indicating moderate predictive performance. While the model captures some patterns in the data, it is not highly reliable and should be used along with other indicators.

Some stocks showed high volatility values, indicating sudden price fluctuations. Additionally, many stocks generated SELL signals, suggesting a bearish trend in the observed period. RSI values initially showed missing values due to insufficient data for calculation.

CHARTS:


Stock Price with Moving Averages
PRICE
1200
1000
800
600
400
200
0
02-05-2024
02-06-2024
02-07-2024
02-08-2024
02-09-2024
02-10-2024
02-11-2024
02-12-2024
02-01-2025
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02-01-2026
02-02-2026
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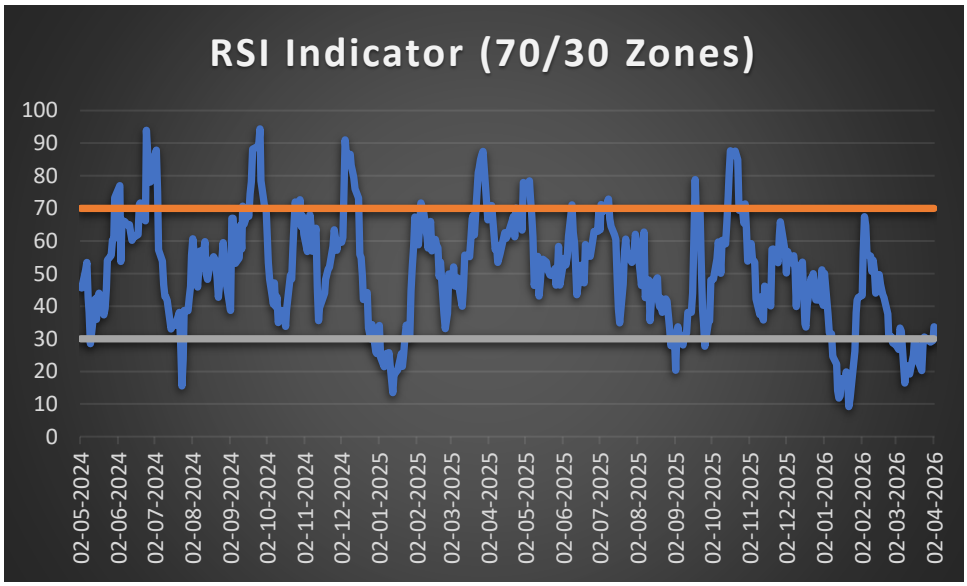
RSI Indicator (70/30 Zones)
100
90
80
70
60
50
40
30
20
10
0
02-05-2024
02-06-2024
02-07-2024
02-08-2024
02-09-2024
02-10-2024
02-11-2024
02-12-2024
02-01-2025
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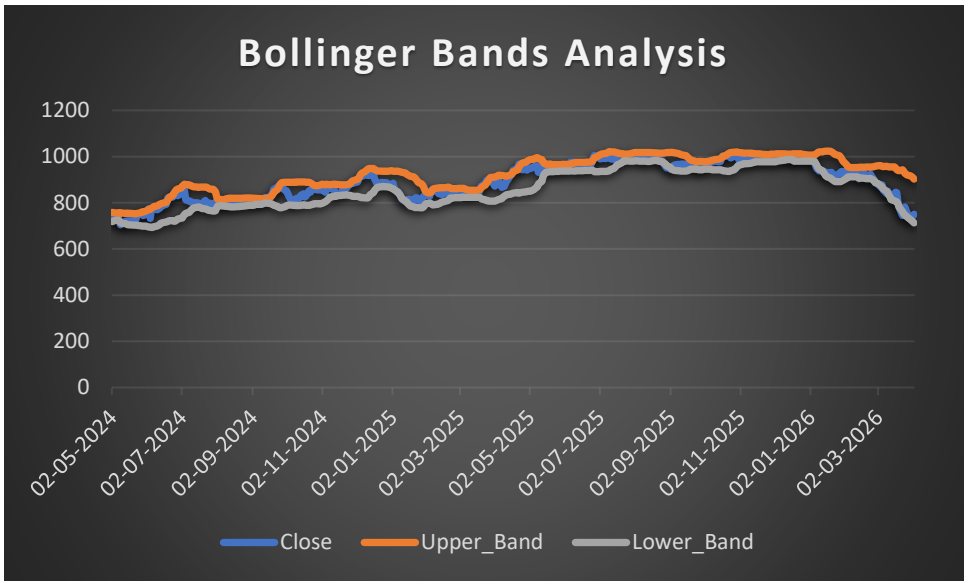
02-04-2026



Ticker	Signal	Volatili	Predict
HDFCBANI	SELL	61.13973	SELL
HINDUNILV	SELL	108.5579	SELL
ICICBANK.	SELL	68.85471	SELL
INFY.NS	SELL	40.20128	SELL
TCS.NS	SELL	118.5779	SELL

Model predicted SELL signals based on recent trends and indicators.

Which means your model predicts current market is bearish

PORTFOLIO RECOMMENDATION:

My suggestion if I had 1,00,000 rupees to invest.

Based on the analysis, a conservative allocation strategy is recommended:

- ₹40,000 in a relatively stable stock (HDFCBANK.NS)
- ₹30,000 in a moderately performing stock (TCS.NS)
- ₹20,000 in a diversified stock (INFY.NS)
- ₹10,000 kept as cash or for high-risk opportunities

This allocation balances risk and return while considering volatility and model predictions.

ASSUMPTIONS & LIMITATIONS:

This analysis is based on historical data, and past performance does not guarantee future results. The machine learning model is limited by the quality and quantity of data. External factors such as market news, economic conditions, and global events are not considered in this model.

CONCLUSION:

This analysis is based on historical data, and past performance does not guarantee future results. The machine learning model is limited by the quality and quantity of data. External factors such as market news, economic conditions, and global events are not considered in this model.